

```
[2]: import numpy as np
import pandas as pd

[3]: df = pd.read_csv('/home/subramanyaprasad/Machine_Learning/Data_Analysis/Notes/03-Pandas/tips.csv')

[4]:

[4]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99  1.01  Female    No  Sun  Dinner     2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34  1.66  Male     No  Sun  Dinner     3         3.45    Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50  Male     No  Sun  Dinner     3         7.00    Travis Walters  6011812121971322  Sun4458
3      23.68  3.31  Male     No  Sun  Dinner     2        11.84  Nathaniel Harris  4676137647685994  Sun5260
4      24.59  3.61  Female    No  Sun  Dinner     4         6.15    Tonya Carter  4832732618637221  Sun2251
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
239    29.03  5.92  Male     No  Sat  Dinner     3         9.68    Michael Avila  5296068606052842  Sat2657
240    27.18  2.00  Female    Yes  Sat  Dinner     2        13.59  Monica Sanders  350680615565404   Sat1766
241    22.67  2.00  Male     Yes  Sat  Dinner     2        11.34    Keith Wong  6011891618747196  Sat3880
242    17.82  1.75  Male     No  Sat  Dinner     2         8.91    Dennis Dixon  4375320550050   Sat17
243    18.78  3.00  Female    No  Thur Dinner     2         9.39    Michelle Hardin  3511451626698139  Thur672

244 rows x 11 columns

[5]: df.describe()

[5]:
total_bill  tip    size  price_per_person  CC Number
count  244.000000  244.000000  244.000000  244.000000  2.440000e+02
mean    19.785943    2.988279    2.598672    7.88197    2.583498e+15
std     8.902412    1.383638    0.951100    2.914234    2.389340e+15
min     10.700000    1.000000    1.000000    2.880000    6.040679e+10
25%    13.347500    2.000000    2.000000    5.800000    3.040731e+13
50%    17.795000    2.900000    2.000000    7.255000    3.525319e+15
75%    24.127500    3.562500    3.000000    9.390000    4.838076e+15
max    50.810000    10.000000    6.000000    20.270000    6.598454e+15

[7]: df.sort_values('tip')

[7]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
92     5.75  1.00  Female    Yes  Fri  Dinner     2         2.88    Leah Ramirez  350891676966392   Fri3780
111    7.25  1.00  Female    No  Sat  Dinner     1         7.25    Terri Jones  3559221007826887  Sat4801
67     3.07  1.00  Female    Yes  Sat  Dinner     1         3.07    Tiffany Brock  4359488526995267  Sat3455
236    12.60  1.00  Male     Yes  Sat  Dinner     2         6.30    Matthew Myers  3543676378973965  Sat5032
0      16.99  1.01  Female    No  Sun  Dinner     2         8.49  Christy Cunningham  3560325168603410  Sun2959
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
141    34.30  6.70  Male     No  Thur  Lunch     6         5.72    Steven Carlson  3526515703718508  Thur1025
59     48.27  6.73  Male     No  Sat  Dinner     4        12.07    Brian Ortiz  6596453823950595  Sat8139
23     39.42  7.58  Male     No  Sat  Dinner     4         9.86    Lance Peterson  3542584061608908  Sat239
212    48.33  9.00  Male     No  Sat  Dinner     4        12.08    Alex Williamson  6762118815212   Sat4590
170    50.81  10.00  Male     Yes  Sat  Dinner     3        16.94    Gregory Clark  5473850968388236  Sat1954

244 rows x 11 columns

Default it's in ascending order but we can also do descending:

[8]: df.sort_values('tip',ascending=False)

[8]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
170    50.81  10.00  Male     Yes  Sat  Dinner     3        16.94    Gregory Clark  5473850968388236  Sat1954
212    48.33  9.00  Male     No  Sat  Dinner     4        12.08    Alex Williamson  6762118815212   Sat4590
23     39.42  7.58  Male     No  Sat  Dinner     4         9.86    Lance Peterson  3542584061608908  Sat239
59     48.27  6.73  Male     No  Sat  Dinner     4        12.07    Brian Ortiz  6596453823950595  Sat8139
141    34.30  6.70  Male     No  Thur  Lunch     6         5.72    Steven Carlson  3526515703718508  Thur1025
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
0      16.99  1.01  Female    No  Sun  Dinner     2         8.49  Christy Cunningham  3560325168603410  Sun2959
111    7.25  1.00  Female    No  Sat  Dinner     1         7.25    Terri Jones  3559221007826887  Sat4801
92     5.75  1.00  Female    Yes  Fri  Dinner     2         2.88    Leah Ramirez  350891676966392   Fri3780
67     3.07  1.00  Female    Yes  Sat  Dinner     1         3.07    Tiffany Brock  4359488526995267  Sat3455
236    12.60  1.00  Male     Yes  Sat  Dinner     2         6.30    Matthew Myers  3543676378973965  Sat5032

244 rows x 11 columns

We can sort by multiple columns:

[9]: df.sort_values(['tip','size'])

[9]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
67     3.07  1.00  Female    Yes  Sat  Dinner     1         3.07    Tiffany Brock  4359488526995267  Sat3455
111    7.25  1.00  Female    No  Sat  Dinner     1         7.25    Terri Jones  3559221007826887  Sat4801
92     5.75  1.00  Female    Yes  Fri  Dinner     2         2.88    Leah Ramirez  350891676966392   Fri3780
236    12.60  1.00  Male     Yes  Sat  Dinner     2         6.30    Matthew Myers  3543676378973965  Sat5032
0      16.99  1.01  Female    No  Sun  Dinner     2         8.49  Christy Cunningham  3560325168603410  Sun2959
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
141    34.30  6.70  Male     No  Thur  Lunch     6         5.72    Steven Carlson  3526515703718508  Thur1025
59     48.27  6.73  Male     No  Sat  Dinner     4        12.07    Brian Ortiz  6596453823950595  Sat8139
23     39.42  7.58  Male     No  Sat  Dinner     4         9.86    Lance Peterson  3542584061608908  Sat239
212    48.33  9.00  Male     No  Sat  Dinner     4        12.08    Alex Williamson  6762118815212   Sat4590
170    50.81  10.00  Male     Yes  Sat  Dinner     3        16.94    Gregory Clark  5473850968388236  Sat1954

244 rows x 11 columns

To get index of max or min values:

[10]: df['total_bill'].idxmax()

[10]: 170

[11]: df['total_bill'].idxmin()

[11]: 67

Entire:

[12]: df.iloc[df['total_bill'].idxmax()]

[12]:
total_bill    50.81
tip           10.0
sex           Male
smoker        Yes
day           Sat
time         Dinner
size           3
price_per_person  16.94
Payer Name    Gregory Clark
CC Number     547385096838236
Payment ID    Sat1954
Name: 170, dtype: object

[14]: df.corr('pearson')

.....
ValueError                                Traceback (most recent call last)
Cell In[14], line 1
----> 1 df.corr('pearson')

File ~/Machine_Learning/Data_Analysis/venv/lib64/python3.14/site-packages/pandas/core/frame.py:1313, in DataFrame.corr(self, method, min_periods, numeric_only)
   1310     ... # noqa: E501
   1318     data = self.get_numeric_data() # numeric_only does self
   1319     cols = data.columns
   1320     idx = cols.columns
   1321     mat = data.to_numpy(dtype=float, na_value=np.nan, copy=False)
-> 1313     # method == 'pearson':
   1314     # cor = libalgos.nancorr(mat, minp=min_periods)
   1315     cor = libalgos.nancorr(mat, minp=min_periods)

File ~/Machine_Learning/Data_Analysis/venv/lib64/python3.14/site-packages/pandas/core/frame.py:2881, in DataFrame.to_numpy(self, dtype, copy, na_value)
   2877     ... [ ..., ..., Timestamp('2000-01-02 00:00:00')]], dtype=object)
   2878     ...
   2879     # dtype = np.dtype(dtype)
   2880     result = self._mgr.as_array(dtype=dtype, copy=copy, na_value=na_value)
-> 2881     # result.dtype
   2882     result = np.asarray(result, dtype=dtype)
   2883     result = np.asarray(result, dtype=dtype)
   2884     result = np.asarray(result, dtype=dtype)

File ~/Machine_Learning/Data_Analysis/venv/lib64/python3.14/site-packages/pandas/core/internals/managers.py:1872, in BlockManager.as_array(self, dtype, copy, na_value)
   1869     ... arr.flags.writeable = False
   1870     ...
-> 1872     arr = self._interleave(dtype=dtype, na_value=na_value)
   1873     # The underlying array was copied without interleaving, so no need
   1874     # to further copy if copy=True or setting na_value
   1875     # na_value is lib.no_default:

File ~/Machine_Learning/Data_Analysis/venv/lib64/python3.14/site-packages/pandas/core/internals/managers.py:1925, in BlockManager._interleave(self, dtype, na_value)
   1911     blk = blk.get_values(dtype)
   1912     blk.is_extension:
   1920     # Avoid implicit conversion of extension blocks to object
   1921     # Avoid implicit conversion of extension blocks to object
   1922     # error: Item "ndarray" of "Union[ndarray, ExtensionArray]" has no
   1923     # attribute "to_numpy"
-> 1925     arr = blk.values.to_numpy() # type: ignore[union-attr]
   1926     # dtype=dtype
   1927     # na_value=na_value
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